

PERSONAL INFORMATION

Born on July 7, 1997 in Rome, Italy.

EDUCATION

Ph.D.	University of Rome Tor Vergata, Ph.D. in Mathematics (European Label), summa cum laude. Thesis: " <i>New ideas for walk-based measures in complex networks</i> ". under the supervision of Prof. Daniele Bertaccini. Defence date: 30th April 2026.	Nov 2022 – Apr 2026
Master degree	University of Rome Tor Vergata, Pure and Applied Mathematics, 110/110 magna cum laude. Thesis: " <i>Robustness and centrality indices in some complex networks</i> ", under the supervision of Prof. Daniele Bertaccini.	Apr 2022
Bachelor degree	University of Rome Tor Vergata, Mathematics, 108/110.	Oct 2019

RESEARCH

Interests: Numerical Linear Algebra * Matrix Theory * Network Science.

Research abroad

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| ✈ Aalto University, Espoo, Finland - under the
supervision of Prof. Vanni Noferini, two weeks. | Sept 2025 |
| ✈ University of Strathclyde, Glasgow, UK - under the
supervision of Dr. Francesca Arrigo, three months. | Nov 2023 - June 2025 |

PUBLICATIONS

Published articles

- D. Bertaccini, L. Chiricosta, and A. Filippo. *Source-to-sink communicability: a new centrality measure for directed acyclic networks*. Numerical Algorithms, Sept. 2025.
- F. Arrigo, D. Bertaccini, and A. Filippo. *Updating Katz centrality by counting walks*. SIAM Journal on Matrix Analysis and Applications, 2025.
- D. Bertaccini and A. Filippo. *A proposal for ranking through selective computation of centrality measures*. PLOS ONE, 2023.

Submitted / in press articles

- A. Filippo and M. Mazza. *Spectral and computational aspects of a regularized fractional Laplacian for non-local diffusion on graphs*. Accepted for publication in Journal of Numerical Mathematics, 2026.

TALKS AND POSTERS

- 👤 Mathematics of Data Driven Models, GSSI, L'Aquila, Italy. **Talk:** "On superdiffusive processes on graphs using a regularized fractional Laplacian". Jan 2026
- 👤 DocTorV Junior Seminars, University of Rome Tor Vergata, Italy. **Talk:** "Walk -based centrality measures in Complex Networks" Oct 2025
- 👤 NA Seminar, Aalto University, Finland. **Talk:** "Walk-induced strategies for updating Katz centrality and applications to Network Robustness". Sept 2025
- 👤 Modern Perspectives in Approximation Theory: Graphs, Networks, quasi-interpolation and Sampling Theory, Cetraro, Italy. **Poster:** "Updating Katz centrality via walk count". July 2025
- 👤 30th Biennial NACONF, University of Strathclyde, Glasgow, UK. **Talk:** "A new walk-based centrality measure for directed acyclic networks". June 2025
- 👤 2ggALN, University of Pisa, Italy. **Talk:** "Walk-induced strategies for updating Katz centrality in simple graphs". Jan 2025
- 👤 CACN24, University of Rome Tor Vergata, Italy. **Poster:** "New numerical strategies for approximately updating Katz centrality". Dec 2024
- 👤 Numerical Linear Algebra, CIRM Luminy, France. **Poster:** "Efficient strategies for updating some resolvent-based centrality measures". Sept 2024
- 👤 PYSANUM Seminar, University of Pisa, Pisa, Italy. **Talk:** "Complex Networks: robustness, centrality indices and new strategies for ranking nodes". Dec 2023
- 👤 NA Seminar, University of Strathclyde, Glasgow, UK. **Talk:** "Node ranking via selective computation of centrality measures". Nov 2023
- 👤 2ggALN, GSSI, L'Aquila, Italy. **Poster:** "A proposal for ranking through selective computation of centrality measures". May 2023

TEACHING

- 👤 **University tutor at Engineering Department**, Geometry, University of Rome Tor Vergata, Rome, Italy. March 2023 - Jan 2026
- 👤 **University tutor at Mathematics Department**, Geometry 1, University of Rome Tor Vergata, Rome, Italy. Oct 2021 - Jan 2022

OTHER SKILLS

- 🌐 **Languages:** Italian (native), English (upper intermediate).
- 💻 **Informatics:** Matlab, Python, C, R, Eiffel.

Last update: 04/05/2026